Ryerson University
Department of Mechanical and Industrial Engineering

MTL 200 – Materials Science Fundamentals

Midterm Test No.1

February 24, 2016

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INSTRUCTIONS

1. Duration is **50 minutes**.
2. This is a closed book test.
3. Back pages may be used if extra space is needed.
5. Please use an ink pen/ballpoint pen, and do not use a pencil (except for drawings if any).
6. Total marks: **40**

LAST NAME: Solution _____ **FIRST NAME:** _____
CAPITAL LETTERS CAPITAL LETTERS

STUDENT NO. _____

SECTION NO. _____

Questions	**1**	**2**	**3**	**4**	**5**	**Total**
Maximum marks	**10**	**6**	**4**	**10**	**10**	**40**
Actual marks						

1. Circle the correct answer for the following questions (10 marks).

(1) The number of energy states for the *d* subshell is

(a) 1
(b) 3
(c) 5
(d) 7

(2) Which electron configuration is stable?

(a) $1s^2 2s^2 2p^6 3s^2 4s^1$
(b) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
(c) $1s^2 2s^2 2p^6 3s^2 3p^6$
(d) $1s^2 2s^2 2p^6 3s^2$

(3) What type(s) of bonding would be expected for ceramics?

(a) Ionic bonding
(b) Covalent bonding
(c) Ionic and Covalent bonding
(c) Covalent and Secondary bonding

(4) The coordination number in the unit cell of hexagonal close-packed (HCP) crystal structure is

(a) 4
(b) 8
(c) 12
(d) 16

(5) The stacking sequence of close-packed planes in the form of ABCABC… is for

(a) FCC
(b) BCC
(c) HCP
(d) none of the above crystal structures

(6) Using X-ray diffraction, it was found that a material had constructive interference for the (311) and (222) planes. What is the crystal structure of this material?

(a) FCC
(b) BCC
(c) HCP
(d) none of the above crystal structures

(7) Four main factors affecting the solubility of the substitutional solid solution are

(a) composition, crystal structure, electronegativity, valence
(b) atomic size, composition, electronegativity, valence
(c) atomic size, crystal structure, composition, valence
(d) atomic size, crystal structure, electronegativity, valence

(8) Which of the following is a linear defect?

(a) A Schottky defect
(b) A screw dislocation
(c) A Frenkel defect
(d) twin boundaries

(9) Diffusion by which mechanism occurs more rapidly in metal alloys?

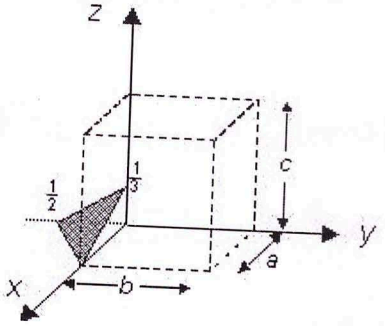
(a) volume diffusion
(b) vacancy diffusion
(c) interstitial diffusion
(d) none of the above

(10) For non-steady state diffusion, concentration of diffusion species is a function of

(a) position only
(b) time only
(c) both
(d) none

2. Short Answer. (Enter the final answer in the box provided. Only the final answer will be graded. Please include units in your answer if necessary)

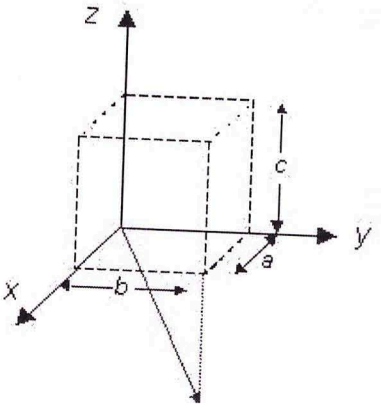
(a) Determine the Miller index for the following crystallographic plane **(2 marks)**.



Answer:

$(1\bar{2}3)$

(b) Determine the Miller index for the following crystallographic direction **(2 marks)**.



Answer:

$[11\bar{1}]$

(c) Name two materials from the following which have covalent bonding **(2 marks)**.

NaCl
MgO
Cl_2
Ag
Hg
C (diamond)
Al
Mg
CaF_2

Answer:

Cl_2

C (diamond)

3. Strontium (Sr) has an FCC crystal structure, an atomic radius of 0.215 nm and an atomic weight of 87.62 g/mol. Calculate the theoretical density for Sr **(4 marks)**.

$R = 0.215\,nm$, $A_{Sr} = 87.62\ g/mol$

FCC, $n = 4$ atoms/unit cell

Cube volume $= V = a^3$, $a = 2\sqrt{2}R$

$$V_c = (2\sqrt{2}R)^3$$

$$\rho = \frac{nA}{V_c N_A}$$

$$= \frac{4 \times 87.62}{2\sqrt{2}\,(0.215 \times 10^{-7}) \times 6.022 \times 10^{23}}$$

$$= 2.59\ g/cm^3$$

4. The metal niobium (Nb) has a BCC crystal structure. If the angle of diffraction for the (211) set of planes occurs at 75.99° (first-order reflection) when monochromatic x-radiation having a wavelength of 0.1659 nm is used, compute the following: (a) the interplanar spacing for this set of planes and (b) the atomic radius for the Nb atom **(10 marks)**.

$2\theta = 75.99^\circ \quad \therefore \theta = 37.995^\circ$

$n = 1 \quad \lambda = 0.1659\ nm$

(1) $n\lambda = 2d \sin\theta$

$(1) \times 0.1659 = 2 \cdot d \cdot \sin 37.995^\circ$

$d = 0.1347\ nm$

(2) $d_{hkl} = \dfrac{a}{\sqrt{h^2+k^2+l^2}}$

$0.1347 = \dfrac{a}{\sqrt{2^2+1^2+1^2}}$

$\therefore a = 0.33\ nm$

for BCC $a = \dfrac{4R}{\sqrt{3}}$

$0.33 = \dfrac{4R}{\sqrt{3}}$

$\therefore R = 0.1429\ nm$

5. An FCC iron-carbon alloy initially containing 0.10 wt% C is carburized at an elevated temperature and in an atmosphere in which the surface carbon concentration is maintained at 1.10 wt%. If after 48 h the concentration of carbon is 0.30 wt% at a position 3.5 mm below the surface, determine the diffusion co-efficient **(10 marks)**.

Soln:

$C_o = 0.10$ wt% $t = 48$ h

$C_s = 1.10$ wt% $C_x = 0.30$ wt% $x = 3.5$ mm

$$\frac{C_x - C_o}{C_s - C_o} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\frac{0.3 - 0.1}{1.1 - 0.1} = 1 - \text{erf}(z)$$

where $z = \frac{x}{2\sqrt{Dt}}$

$$0.2 = 1 - \text{erf}(z)$$

$\therefore$ erf(z) = 0.8

erf(z)	z
0.7970	0.90
0.80	z
0.8290	0.95

By taking iteration

$$\frac{z - 0.9}{0.8 - 0.7970} = \frac{0.95 - 0.9}{0.8290 - 0.7970}$$

$\therefore z = 0.9047$

$$\frac{x}{2\sqrt{Dt}} = 0.9047$$

$$\frac{3.5 \times 10^{-3}\,m}{2\sqrt{D} \times \sqrt{48 \times 3600}} = 0.9047$$

$\therefore D = 2.16 \times 10^{-11}\ m^2/s$

Equations and Constants:

$n\lambda = 2d_{hkl}\sin\theta$ $\quad d_{hkl} = \dfrac{a}{\sqrt{h^2+k^2+l^2}}$ $\quad N_M\left(\dfrac{M}{100}\right)^2 = 2^{n-1}$ $\quad A_{ave} = \dfrac{100}{\dfrac{C_1}{A_1}+\dfrac{C_2}{A_2}}$

$\rho_{ave} = \dfrac{100}{\dfrac{C_1}{\rho_1}+\dfrac{C_2}{\rho_2}}$ $\quad \rho = \dfrac{n\ A}{V_c\ N_A}$ $\quad N = 2^{n-1}$ $\quad N_v = N\exp\left(-\dfrac{Q_v}{kT}\right)$

$\mathrm{J} = -D\dfrac{\mathrm{dC}}{\mathrm{dx}}$ $\quad J = \dfrac{M}{At}$ $\quad D = D_o\exp\left(-\dfrac{Q_d}{RT}\right)$ $\quad \dfrac{C_x - C_o}{C_s - C_o} = 1 - erf\left(\dfrac{x}{2\sqrt{Dt}}\right)$

FCC: $a = 2\sqrt{2}R$ $\quad$ BCC: $a = \dfrac{4R}{\sqrt{3}}$ $\quad$ HCP: $a = 2R$ $\quad$ Electron charge $e = 1.602\times10^{-19}$ C

Electron mass = 9.11×10^{-31} kg
Gas constant R = 8.31 J/mol-K

Boltzmann's constant $k = 8.62\times10^{-5}$ eV/atom-K
Avogadro's number $N_A = 6.023\times10^{23}$ molecules/mol

z	*erf(z)*	*z*	*erf(z)*	*z*	*erf(z)*
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999